双末端闭式循环・公开教学示例・计算报告

OpenHydroNet 0.1.0-preview.4 | 2026-10-27T12:13:30.458599+00:00

状态：converged；算法：sparse-head-flow-newton-trf-v2

物理输入 SHA-256：f6e62dbcd139f0ad090e2fa44056deb4eb3a0fa9bb4957c7f9183f04237de623

假设和验证


```
{
  "assumptions": [
    "稳态不可压单相液态水",
    "节点压力采用表压，H=z+p/(rho*g)",
    "不含水锤、CFD 或时间状态"
  ],
  "fluid": {
    "density_kg_m3": 998.2336361398824,
    "kinematic_viscosity_m2_s": 1.0035213432425121e-06,
    "temperature_c": 20
  },
  "residuals": {
    "mass_m3_s": 0,
    "energy_m": 1.1479812656034483e-09,
    "mass_tolerance_m3_s": 1e-08,
    "energy_tolerance_m": 1e-05
  },
  "issues": [
    {
      "code": "TERMINAL_DEMAND",
      "object_id": "T2",
      "message": "末端流量低于设计需求，固定工况不满足",
      "severity": "warning"
    }
  ]
}
```


支路结果

设备 ID	名称	类型	流量 [m³/h]	质量流量 [kg/h]	流速 [m/s]	测压水头差 [m]	水力功率 [W]	估算输入功率 [W]
P1	P1	pipe	31.17815	31123.08	1.722971	3.126067	—	—
P2	P2	pipe	13.6786	13654.44	1.145045	2.382101	—	—
P3	P3	pipe	13.6786	13654.44	1.145045	2.382101	—	—

设备 ID	名称	类型	流量 [m³/h]	质量流量 [kg/h]	流速 [m/s]	测压水头差 [m]	水力功率 [W]	估算输入功率 [W]
P4	P4	pipe	31.17815	31123.08	1.722971	3.126067	—	—
PU1	循环泵 · 虚构示例	pump	31.17815	31123.08	—	-18.49893	1568.366	—
V1	调节阀 1	valve	17.49955	17468.64	—	0.4322102	—	—
T1	用户末端 1	terminal	17.49955	17468.64	—	11.81459	—	—
V2	调节阀 2	valve	13.6786	13654.44	—	0.2640735	—	—
T2	用户末端 2	terminal	13.6786	13654.44	—	7.218523	—	—

无效率数据时不提供输入功率；水力功率不等于整机电功率。压力水头不等于总水头。

完整输入（SI）


```
{
  "nodes": [
    {
      "id": "s0",
      "name": "泵出口",
      "elevation": 0,
      "demand": 0,
      "head": null
    },
    {
      "id": "s1",
      "name": "供水分配",
      "elevation": 0,
      "demand": 0,
      "head": null
    },
    {
      "id": "s2",
      "name": "远端供水",
      "elevation": 0,
      "demand": 0,
      "head": null
    },
    {
      "id": "t1",
      "name": "末端一入口",
      "elevation": 0,
      "demand": 0,
      "head": null
    },
    {
      "id": "t2",
      "name": "末端二入口",
      "elevation": 0,
      "demand": 0,
      "head": null
    },
    {
```

```
    "id": "r0",
    "name": "定压点 / 泵入口",
    "elevation": 0,
    "demand": 0,
    "head": 30
  },
  {
    "id": "r1",
    "name": "回水汇合",
    "elevation": 0,
    "demand": 0,
    "head": null
  },
  {
    "id": "r2",
    "name": "远端回水",
    "elevation": 0,
    "demand": 0,
    "head": null
  }
],
"edges": [
  {
    "id": "P1",
    "name": "P1",
    "kind": "pipe",
    "source": "s0",
    "target": "s1",
    "length": 75,
    "diameter": 0.08,
    "roughness": 5e-05,
    "zeta": 2,
    "local_mode": "zeta"
  },
  {
    "id": "P2",
    "name": "P2",
    "kind": "pipe",
    "source": "s1",
    "target": "s2",
    "length": 100,
    "diameter": 0.065,
    "roughness": 5e-05,
    "zeta": 2,
    "local_mode": "zeta"
  },
  {
    "id": "P3",
    "name": "P3",
    "kind": "pipe",
    "source": "r2",
    "target": "r1",
    "length": 100,
    "diameter": 0.065,
    "roughness": 5e-05,
    "zeta": 2,
    "local_mode": "zeta"
  },
  {
    "id": "P4",
    "name": "P4",
    "kind": "pipe",
    "source": "r1",
    "target": "r0",
    "length": 75,
```

```
    "diameter": 0.08,
    "roughness": 5e-05,
    "zeta": 2,
    "local_mode": "zeta"
  },
  {
    "id": "PU1",
    "name": "循环泵 · 虚构示例",
    "kind": "pump",
    "source": "r0",
    "target": "s0",
    "curve": [
      32,
      0,
      -180000
    ],
    "speed": 1,
    "parallel": 1,
    "check_valve": true,
    "valid_flow": [
      0,
      0.013
    ],
    "data_source": "人工构造，非厂家数据"
  },
  {
    "id": "V1",
    "name": "调节阀 1",
    "kind": "valve",
    "source": "s1",
    "target": "t1",
    "opening": 1,
    "kv": 85,
    "characteristic": "linear"
  },
  {
    "id": "T1",
    "name": "用户末端 1",
    "kind": "terminal",
    "source": "t1",
    "target": "r1",
    "design_head": 8,
    "design_flow": 0.004
  },
  {
    "id": "V2",
    "name": "调节阀 2",
    "kind": "valve",
    "source": "s2",
    "target": "t2",
    "opening": 1,
    "kv": 85,
    "characteristic": "linear"
  },
  {
    "id": "T2",
    "name": "用户末端 2",
    "kind": "terminal",
    "source": "t2",
    "target": "r2",
    "design_head": 8,
    "design_flow": 0.004
  }
],
"fluid": {
```

```
    "temperature": 20,
    "atmospheric_pressure": 101325
  }
}
```